

Estructura de Datos

Amy Cárdenas Silva

Grupo: 1360

Carrera: ingeniería en
computación

tarea 12, muestra de
funcionamiento

```

import java.util.ArrayList;
import java.util.List;

public class Main {
    public static ArrayList mergeSort(ArrayList data) {
        if(data.size() > 1){
            int mitad = data.size() / 2;
            ArrayList<Integer> izq = new ArrayList<>();
            ArrayList<Integer> der = new ArrayList<>();
            for (int i = 0; i < mitad; i++) {
                izq.add((int) data.get(i));
            }
            int restante = data.size() - mitad;
            for (int i = 0; i < restante; i++) {
                der.add((int) data.get(i + mitad));
            }
            System.out.println(izq + " " + der);
            mergeSort(izq);
            mergeSort(der);
            int i = 0;
            int d = 0;
            int k = 0;
            while (i < izq.size() && d < der.size()) {
                if (izq.get(i) < der.get(d)) {
                    data.set(k, izq.get(i));
                    i++;
                } else {
                    data.set(k, der.get(d));
                    d++;
                }
                k++;
            }
            while (i < izq.size()){
                data.set(k, izq.get(i));
                i++;
                k++;
            }
            while (d < der.size()){
                data.set(k, der.get(d));
                d++;
                k++;
            }
        }
        System.out.println("regreso de rec :" + data);
        return data;
    }

    public static void main(String[] args) {
        ArrayList info = new ArrayList<>();
        List<Integer> lista = List.of(1,11,2,10,3,9,4,8,5,7,6);
        for (Integer item: lista){
            info.add(item);
        }
    }
}

```

```
mergeSort(info);  
}  
}
```

```
[1, 11, 2, 10, 3] [9, 4, 8, 5, 7, 6]  
[1, 11] [2, 10, 3]  
[1] [11]  
regreso de rec :[1]  
regreso de rec :[11]  
regreso de rec :[1, 11]  
[2] [10, 3]  
regreso de rec :[2]  
[10] [3]  
regreso de rec :[10]  
regreso de rec :[3]  
regreso de rec :[3, 10]  
regreso de rec :[2, 3, 10]  
regreso de rec :[1, 2, 3, 10, 11]  
[9, 4, 8] [5, 7, 6]  
[9] [4, 8]  
regreso de rec :[9]  
[4] [8]  
regreso de rec :[4]  
regreso de rec :[8]  
regreso de rec :[4, 8]  
regreso de rec :[4, 8, 9]  
[5] [7, 6]  
regreso de rec :[5]  
[7] [6]
```

```
regreso de rec :[7]  
regreso de rec :[6]  
regreso de rec :[6, 7]  
regreso de rec :[5, 6, 7]  
regreso de rec :[4, 5, 6, 7, 8, 9]  
regreso de rec :[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
```

Process finished with exit code 0